

Rafael Reis

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EDUCATION

McGill University	Montreal, QC
<i>Master of Engineering in Applied Artificial Intelligence</i>	<i>Aug. 2026 – Exp. May 2028</i>
McGill University	Montreal, QC
<i>Bachelor of Engineering in Computer Engineering, Minor in Applied Artificial Intelligence</i>	<i>Sep. 2021 – May 2026</i>

EXPERIENCE

Undergraduate Research Assistant for Dr. Derek Ruths	Jan 2026 – May 2026
<i>Network Dynamics Lab / McGill University</i>	<i>Montreal, QC</i>
• Built a reproducible counterspeech generation pipeline (OpenAI Responses API, GPT-4.1-mini) with retry/backoff and resumable runs to compare a base model against failure-mode fine-tuned variants from the lab's LLM-as-a-judge study • Curated a seeded, target-balanced evaluation set of 100 hate speech examples (20 per group across 5 target groups) from 596 MT-CONAN items • Designed a blinded pairwise human evaluation protocol (randomized A/B presentation, preference plus 1–5 degree-of-difference) with a per-group, paired-test analysis plan	
Undergraduate Research Assistant for Dr. Amir Shmuel	May 2025 – Aug 2025
<i>The Neuro / McGill University</i>	<i>Montreal, QC</i>
• Implemented MATLAB/Python data pipelines for Optical Imaging, reducing analysis time and improving reproducibility • Developed scripts to extract and quantify neural activity from fMRI and Optical Imaging datasets for lab research • Designed and 3D-printed custom imaging fixtures in Blender and UltiMaker Cura to support functional imaging experiments	
DevSecOps & Cloud Security Intern	May 2024 – Aug 2024
<i>Canadian National Railway / CISO</i>	<i>Montreal, QC</i>
• Designed and deployed CI/CD pipelines in Azure Pipelines to automate build, test, and deployment for backend security systems using Python, Terraform, YAML, and PowerShell • Wrote Terraform IaC to provision and maintain cloud resources, reducing manual configuration errors across environments • Hardened code infrastructure supporting CN's enterprise security network, reducing attack surface	
Server & Bartender	May 2026 – Present
<i>Breizh Cafe</i>	<i>Montreal, QC</i>
• Independently ran front-of-house for a 41-seat restaurant, averaging 50 covers/weekday shift solo and 90/weekend shift, handling opening/closing and cash flow	
Academic Tutor	2018 – Present
<i>McGill University & Self-Employed</i>	<i>Montreal & Paris</i>
• Tutored university students at McGill in Calculus II, Linear Algebra, Differential Equations and intro to Python; delivered weekly lessons to middle and high school students in France	
IT Risk Management Intern	May 2022 – June 2022
<i>Groupe BCPE & Natiris</i>	<i>Paris, France</i>
• Tracked budgets in Excel and Python across two large-scale IT projects run by five branches, supporting resource allocation and future planning	

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C#, JavaScript, SQL, HTML/CSS, ARM Assembly, VHDL
ML & Frameworks: PyTorch, HuggingFace, Scikit-Learn, NumPy, Pandas, Gym, OpenAI API, React, Unity, Pygame
Reinforcement Learning: PPO, GAE, DQN, MCTS, self-play
Developer Tools: Git, Docker, Terraform, YAML, AWS, Azure DevOps, pytest, VS Code, IntelliJ
Languages: French (native), German (native), English (fluent)

PROJECTS

- Domibot – Deep RL for Dominion** | *Python, PyTorch, PPO, GAE, MCTS, pytest* Aug 2026 – Present
- Built a full Dominion engine (all 26 base-set cards, 2–4 players) with generator-based card effects and a bounded 206-action space for self-play RL; ~10.7k lines of Python, 167 tests
 - Diagnosed why AlphaZero-style MCTS self-play plateaued (68% vs. BigMoney, no multi-card strategy) as a credit-assignment failure after five targeted fixes
 - Switched to PPO + GAE with batched rollouts: ~250x faster per iteration, 95–100% win rate vs. BigMoney, and learned multi-action engine turns
 - Built a live-game move advisor that reconstructs dominion.games positions from parsed logs, determinizes hidden cards, and runs MCTS on the PPO network
- Deep RL for Chess (AlphaZero-inspired)** | *Python, PyTorch, MCTS, CNN* May 2025 – Present
- Implemented self-play RL chess AI (AlphaZero) with a residual CNN predicting policy and value, guiding Monte Carlo Tree Search for move selection
 - GPU-accelerated training and multiprocess self-play; built a human vs. AI interface and tracked performance across iterations
- Deep RL for Settlers of Catan** | *Python, PyTorch, Gym, PPO, DQN* Jan 2025 – Apr 2025
- Trained PPO and DQN agents with custom architectures on a Gym-wrapped Catan simulator’s partially observable, large action space
 - Designed league-style self-play evaluation, achieving a 45% win rate with PPO agents over recent versions; PPO proved more stable, with trading and long-horizon play as open challenges
- Media Coverage Analysis Pipeline** | *Python, NLP, TF-IDF, APIs* Nov 2025 – Dec 2025
- Built a modular pipeline to collect, clean, and analyze North American news coverage of political figures via TheNewsAPI, with TSV workflows for qualitative coding and sentiment annotation
 - Implemented TF-IDF topic characterization and automated reporting to quantify sentiment, topic distributions, and media source biases
- Robe&Hood – Indie Video Game** | *C#, Unity, A** May 2023 – Present
- Developed a 2D stealth-action game in Unity with A* pathfinding and FSM-based enemy AI, plus reusable C# gameplay systems for movement, combat, and scene management

VOLUNTEERING / EXTRACURRICULARS

- Chess Club VP Internal** Sep 2023 – Present
McGill Student’s Chess Club *Montreal, QC*
- Organized weekly meetings and tournaments; on the Executive Committee, contributed to constitutional decisions and funding applications, and supported the Canadian University Chess Championship (Montreal, Jan 2025)
- GameDev McGill Member** Jan 2024 – Present
GameDev McGill *Montreal, QC*
- Won 3rd place at McGameJam 2024 (48h game development hackathon) with ‘Rube’s Rocket Rumble’
- TechFair Volunteer** Oct 2022, Feb 2024, Oct 2024
McGill TechFair *Montreal, QC*
- Volunteered three times with a 50+ student team to set up the event and assist company representatives